SHIWEI LIU

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EDUCATION

Fudan University, Shanghai, China **09/2021 - 06/2024**
Ph.D, Electronic Engineering (GPA: 3.65/4.0, Rank: Top-5%)
- Frontier Institute of Chip and System
- Department of Micro-electronics

Fudan University, Shanghai, China **09/2018 - 06/2021**
M.Eng, Electronic Engineering (GPA: 3.83/4.0, Rank: Top-1%)
- Department of Micro-electronics

Fudan University, Shanghai, China **09/2014 - 06/2018**
B.Eng, Electronic Engineering (GPA: 3.67/4.0, Rank: Top-5%)
- Department of Micro-electronics

RESEARCH INTERESTS AND STRENGTHS

Google Scholar
- https://scholar.google.com/citations?user=uxe2gOAAAAAJ&hl=zh-CN&oi=ao

Efficient Machine Learning
- Efficient AI Computing with Model Compression, Pruning and Quantization
- Finetuning-Free Compression for Large Language Model and Generative AI

Deep Learning Processor Architecture and Chip Designs
- Software-Hardware Co-Design for Efficient LLM Acceleration
- Chiplet-Based Scalable System/Dataflow Design
- Compute-in-Memory Architecture and Circuit Design for Neural Network and LLM
- Custom Instruction Set Design and Workflow Optimization

WORK EXPERIENCE

Research Fellow, University of Michigan, Ann Arbor, MI **07/01/2024 - Now**
Visiting Researcher, University of Michigan, Ann Arbor, MI **11/30/2023 - 07/01/2024**
- **Host:** Mehdi Saligane (mehdi@umich.edu),
- Digital Reconfigurable Architecture and Custom Circuit Design for Neural Network and LLM.
- SRAM-/ReRAM-Based Compute-in-Memory Architecture and Circuit Design for Neural Network and LLM.

Ph.D Research Intern, Google, Mountain View, CA **16/09/2023 - 05/01/2024**
- **Host:** Gregory Kielian (gkielian@google.com), Bangfei Pan (pbf@google.com)
- Hardware-Software Co-Design for efficient LLM Acceleration with Model Compression, Pruning and Quantization.
- Digital Reconfigurable Architecture for Neural Network and LLM.
- Implementation of the Proposed Hardware with Open-Source OpenRoad Tool Chain.

TAPEOUTS

Reconfigurable TPU for Group-Query Attention **Intel 16nm CMOS**
- **Period and Location:** 2024.04-2024.10, University of Michigan and Google Research, US
- **Abstract:** This tapeout is a reconfigurable TPU-like accelerator for GPT and Llama acceleration. The scalability enables efficient continuous batch processing under multi-users-various-sequences scenarios. The reconfigurable TPU core reduces scatter-and-reduce communication overhead in group-query attention. Finally, it utilizes an out-of-order computing scheme to avoid the data dependency of softmax and normalization during LLM decoding.
- **Contributions:** Project Leader, from Architecture Design to Verilog Coding, Physical Implementation (Synthesis and P&R) and test/simulation/demo platform setup.

Computing-in-Memory Accelerator for Sparse Transformer **TSMC 28nm CMOS**

- **Period and Location:** 2022.01-2022.10, Fudan University, Shanghai, China
- **Abstract:** This tapeout is a sparsity-aware compute-in-memory processor for Transformer acceleration. It exploits an semi-unstructured weight sparsity and adaptive local attention in the structured CIM array. This work has been published in the 2023 International Solid-State Circuits Conference (ISSCC).
- **Contributions:** Project Leader, from Architecture Design to Verilog Coding, Physical Implementation (Synthesis and P&R) and test/simulation/demo platform setup.

Multiplier-Free Configurable Accelerator for Low-Bitwidth DNNs **TSMC 65nm CMOS**

- **Period and Location:** 2021.01-2021.08, Fudan University, Shanghai, China
- **Abstract:** This tapeout proposes an in-memory entry-counting scheme to perform low-bitwidth (3/4-bit) quantized DNNs. This scheme exploits sparsity in both activations and weights to avoid multiplication requirement. In addition, this reconfigurable accelerators can support flexible computing flow to fit diverse DNN workload.
- **Contributions:** Co-Author, Synthesis and P&R Implementation and Verification.

PUBLICATIONS

- **Shiwei Liu,** et.al., "McPAL: Scaling Unstructured Sparse Inference with Multi-Chiplet HBM-PIM Architecture for LLMs" in ACM/IEEE Design Automation Conference (DAC), 2025.
- **Shiwei Liu**, et.al., "ConSmax: Hardware-Friendly Alternative Softmax with Learnable Parameters" in ACM/IEEE International Conference on Computer-Aided Design (ICCAD), 2024.
- **Shiwei Liu**, et.al., "HARDSEA: Hybrid Analog-ReRAM and Digital-SRAM In-Memory Computing Accelerator for Dynamic Sparse Self-Attention in Transformer" in IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 2024.
- **Shiwei Liu**, et.al., "A 28nm 53.8TOPS/W 8b Sparse Transformer Accelerator with In-Memory Butterfly Zero Skipper for Unstructured Pruned NN and CIM-Based Local-Attention-Reusable Engine" in IEEE International Solid- State Circuits Conference (ISSCC), 2023.
- **Shiwei Liu**, et.al., "Systolic-Array Deep-Learning Acceleration Exploring Pattern-Indexed Coordinate-Assisted Sparsity for Real-Time On-Device Speech Processing" in Great Lakes Symposium on VLSI (GLSVLSI), 2021.
- **Shiwei Liu**, et.al., "XNORAM: An Efficient Computing-in-Memory Architecture for Binary Convolutional Neural Networks with Flexible Dataflow Mapping" in IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS), 2020.
- Zhirui Huang, **Shiwei Liu**, et.al., "DRAFT: Decoupling Backpropagation from Pre-trained Backbone for Efficient Transformer Fine-Tuning on Edge" in ACM/IEEE Design Automation Conference (DAC), 2025.
- Jie Liao, Bo Jiao, **Shiwei Liu**, et.al., "A Scalable Die-to-Die Interconnect with Replay and Repair Schemes for 2.5D/3D Integration" in IEEE International Symposium on Circuits and Systems (ISCAS), 2023.
- Chen Mu, Yunzhengmao Wang, Jiapei Zheng, **Shiwei Liu**, et.al., "A 200M-Query-Vector/s Computing-in-RRAM ADC-less k-Nearest-Neighbor Accelerator with Time-Domain Winner-Takes-All Circuits" in IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS), 2022.
- Haozhe Zhu, Chixiao Chen, **Shiwei Liu**, et.al., "A Communication-Aware DNN Accelerator on ImageNet Using In-Memory Entry-Counting Based Algorithm-Circuit-Architecture Co-Design in 65-nm CMOS" in IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS), 2020.
- **5 Granted Chinese Patents and 1 US Patent (Pending).**

SELECTED HONORS AND AWARDS

- National Scholarship, 2023 (Top-1%)
- Rong Chang-Fudan Scholarship (Research Award) First Prize, 2023 (Top-1%)
- First Prize of the Scholarship for Outstanding Student of Doctor's Degree, 2022 (Top-5%)
- Awardee of the Outstanding Graduate, Fudan University, 2021, (Top-1%)

VOLUNTEERING

- Invited Reviewer for TCAS-I, TCAS-II, ICLR, TVLSI, OJCAS, TETCI, Expert Systems with Applications, Analog IC and Signal Processing.